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Unit 3 : Session 11 : SLO 1 — IEEE 754 Single Precision Format

■ Section A: Match the Following (Remember)

- A. Sign Bit → 2. Indicates positive or negative number
- B. Exponent Bias → 3. 127 in single precision
- C. Fraction (Mantissa) → 1. 23 bits used for precision representation
- D. Total bits in IEEE 754 (Single) → 5. 32 bits total
- E. Exponent Field → 4. Converts decimal exponent to binary

■ Format Summary

- IEEE 754 single precision = 32 bits: 1 sign bit (bit 31), 8 exponent bits with bias 127 (bits 30-23), 23 mantissa/fraction bits (bits 22-0).
- Value = $(-1)^{\text{sign}} \times 1.\text{mantissa} \times 2^{(\text{exponent} - 127)}$ for normalized numbers.